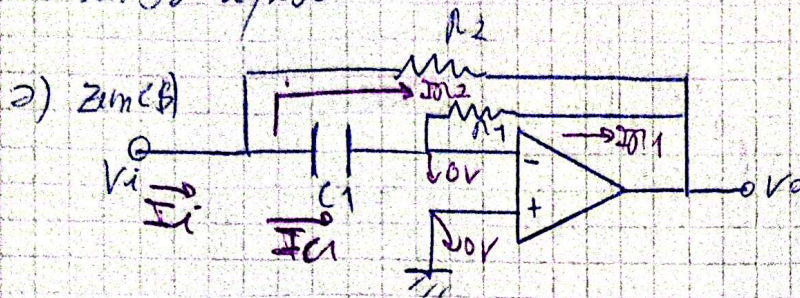


Ejercicio 2) Se desea obtener la impedancia de entrada usando loop



$$Z_e(s) = \frac{1}{sC}$$

$$Z_R(s) = R$$

$V^- = V^+$ por configuración de loop. Amplificador

$$Z_{in}(s) = \frac{V_{in}(s)}{I_{in}(s)}, \quad I_{in}(s) = I_1(s) + I_2(s)$$

$$\text{con } I_1(s) = I_{C1}(s) = I_{R1}(s), \quad I_2(s) = I_{R2}(s)$$

$$\Rightarrow I_{in}(s) = \frac{V_{in}(s) - V^-}{Z_{C1}(s)} = \frac{V^- - V_o(s)}{Z_{R1}(s)} \Rightarrow \frac{V_{in}(s) - 0V}{\frac{1}{sC_1}} = \frac{0V - V_o(s)}{R_1}$$

$$\Rightarrow V_{in}(s) \cdot sC_1 = \frac{-V_o(s)}{R_1} \Rightarrow V_{in}(s) = \frac{-V_o(s)}{sC_1 \cdot R_1} \quad \text{or} \quad \frac{V_o(s)}{V_{in}(s)} = -sC_1 \cdot R_1$$

$$\text{con } A_v(s) = \frac{V_o(s)}{V_{in}(s)} \quad \gamma \quad V_o(s) = -V_{in}(s) \cdot sC_1 \cdot R_1$$

$$I_2(s) = \frac{V_{in}(s) - V_o(s)}{Z_{R2}(s)} = \frac{V_{in}(s) - V_o(s)}{R_2}$$

$$I_{in}(s) = I_1(s) + I_2(s) = \frac{V_{in}(s) - V^-}{Z_{C1}(s)} + \frac{V_{in}(s) - V_o(s)}{R_2}$$

$$I_{in}(s) = \frac{V_{in}(s) - 0V}{\frac{1}{sC_1}} + \frac{V_{in}(s) - (-V_{in}(s) \cdot sC_1 \cdot R_1)}{R_2}$$

$$I_{in}(s) = V_{in}(s) \cdot sC_1 + \frac{V_{in}(s)}{R_2} + V_{in}(s) \cdot \frac{sC_1 \cdot R_1}{R_2}$$

$$I_{in}(s) = V_{in}(s) \cdot \left(sC_1 + \frac{1}{R_2} + \frac{sC_1 \cdot R_1}{R_2} \right)$$

$$\text{Finalmente } \Rightarrow Z_{in}(s) = \frac{V_{in}(s)}{I_{in}(s)} = \frac{V_{in}(s)}{V_{in}(s) \cdot \left(sC_1 + \frac{1}{R_2} + \frac{sC_1 \cdot R_1}{R_2} \right)} \Rightarrow$$

$$Z_m(B) = \frac{1}{\$C_1 + \frac{1}{R_2} + \frac{\$C_1 \cdot R_1}{R_2}} \cdot R_2 = \frac{R_2}{\$C_1 \cdot (R_2 + R_1) + 1} = Z_m(B)$$